

$$\begin{array}{l} \text{Row 3} = \text{Row 2} \\ \times (-1/2) \end{array}$$

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 5 & 5 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 1} = \text{Row 1} \\ -\text{Row 3} \end{array}$$

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 5 & 5 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 2} = \text{Row 2} \\ -5 \times \text{Row 3} \end{array}$$

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 5 & 0 & -4 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{array}{l} \text{Row 1} = \text{Row 1} \\ -\text{Row 2} \times \frac{2}{5} \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & 3/5 \\ 0 & 5 & 0 & -4 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

The corresponding equations are: $x_1 = 3/5$, $5x_2 = -4$, $x_3 = 1$

$$\therefore x_1 = 3/5, x_2 = -4/5, x_3 = 1$$

$$x = \begin{bmatrix} 3/5 \\ -4/5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ -3 & -1 & 2 \\ 0 & 5 & 3 \end{bmatrix} \begin{bmatrix} 3/5 \\ -4/5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$\frac{3}{5} \begin{bmatrix} 1 \\ -3 \\ 0 \end{bmatrix} + (-4/5) \begin{bmatrix} 2 \\ -1 \\ 5 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \rightarrow \text{By defn. of } Ax$$

13. Let $u = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix}$ and $A = \begin{bmatrix} 3 & -5 \\ -2 & 6 \\ 1 & 1 \end{bmatrix}$. Is u in the plane $\mathbb{R}^3$ spanned by the columns of A ? Why or why not?

Solution: By the defn. of span, u is in the Span of the columns of A if u is a linear combination of the columns of A , i.e. $x_1 a_1 + x_2 a_2 = u$, for some $x_1, x_2 \in \mathbb{R}$. This is equivalent to claiming that the corresponding system of equations of the augmented matrix $[a_1 \ a_2 \ u]$ has a solution set.

$$\begin{bmatrix} 3 & -5 & 0 \\ -2 & 6 & 4 \\ 1 & 1 & 4 \end{bmatrix} \xrightarrow[\text{+ Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & -5 & 0 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{bmatrix} \xrightarrow[\text{- Row 2}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 3 & -5 & 0 \\ 1 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow[\text{- } \frac{1}{3} \times \text{Row 1}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & -5 & 0 \\ 0 & 8/3 & 4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times (1/4)]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 3 & -5 & 0 \\ 0 & 2/3 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$